

Solution Architecture

(Automated Network Request Management)

Date	26 September 2026
Team ID	_____
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

1. Introduction

Solution architecture bridges the gap between a business problem and its technology solution. For Automated Network Request Management, it defines how users, ServiceNow interfaces, workflow automation, business logic, data storage, integrations, reporting and infrastructure work together to manage network requests from submission to closure.

2. Objectives of the Solution Architecture

Provide a clear technical structure for the proposed solution.

Map business requirements to ServiceNow platform components.

Define request, approval, fulfillment and status data flows.

Support security, auditability, scalability and maintainability.

Provide extension points for future enterprise and network integrations.

3. High-Level Solution Architecture

The following diagram presents the proposed architecture of the Automated Network Request Management solution.

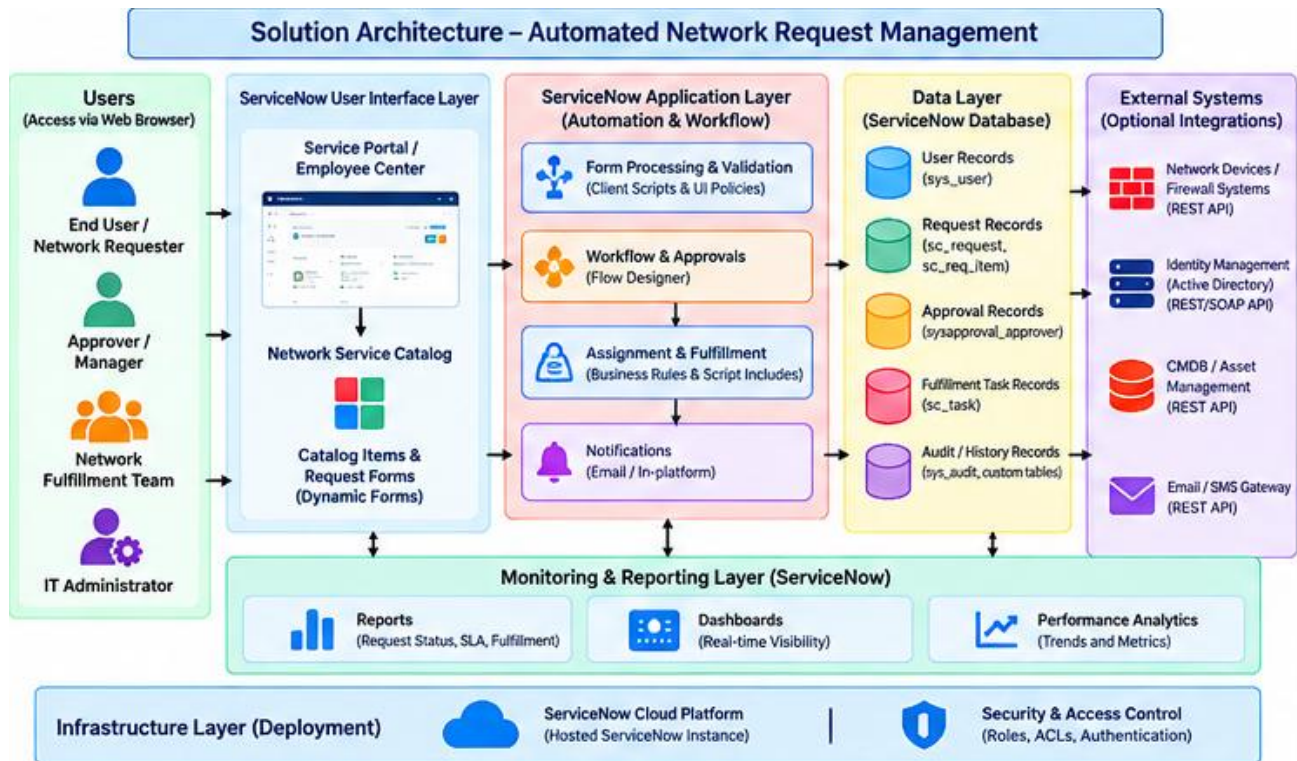


Figure 1: High-Level Solution Architecture

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4. Architecture Description

4.1 User Layer

Includes End Users/Network Requesters, Approvers/Managers, the Network Fulfillment Team and IT Administrators. Each role accesses only the functions required for its responsibilities.

4.2 User Interface Layer

ServiceNow Employee Center or Service Portal provides the user interface. The Network Service Catalog and catalog items provide standardized request forms.

4.3 Application Layer – Automation & Workflow

Client Scripts and UI Policies support form behavior and validation. Flow Designer automates approvals, assignment, fulfillment tasks and notifications. Business Rules and Script Includes support server-side processing and reusable logic.

4.4 Data Layer

The ServiceNow platform database stores user, network request, approval, fulfillment task and audit/history information.

4.5 External Systems

Optional REST/SOAP integrations can connect to network devices, firewall systems, identity management, CMDB/asset systems or notification services.

4.6 Monitoring & Reporting

Reports and dashboards provide visibility into request status, SLA information, fulfillment activity and operational trends.

4.7 Infrastructure & Security

The solution is hosted on the ServiceNow cloud platform. Roles, ACLs, authentication and authorization provide controlled access and auditability.

5. Architecture Components and Functions

Layer / Component	Key Functions
Users	Submit requests, approve requests, perform fulfillment and manage configurations.
Service Portal / Employee Center	Provides access to the service catalog and request forms.
Service Catalog	Provides standardized network request categories and catalog items.
Client Scripts & UI Policies	Control form behavior, mandatory fields and dynamic fields.
Flow Designer	Automates approval routing, assignment, task creation and notifications.
Business Rules & Script Includes	Provide server-side processing and reusable business logic.
ServiceNow Database	Stores users, requests, approvals, tasks and audit information.
External Integrations	Optional integration with network, identity, CMDB/asset or notification systems.
Reports & Dashboards	Provide request, SLA, fulfillment and performance visibility.
Security	Roles, ACLs, authentication and authorization protect access.
Infrastructure	ServiceNow cloud platform hosts the proposed solution.

6. Request Processing Flow

Requester → Service Catalog → Network Request Form → Validation → Approval → Assignment → Network Fulfillment → Status Update → Notification → Closure → Audit / Reporting.

7. Architectural Benefits

Standardized network request intake.

Reduced manual coordination through workflow automation.

Role-based access and auditability.

Improved request visibility and reporting.

Scalable addition of request types, workflows and integrations.

8. Conclusion

The proposed architecture uses ServiceNow as the central platform for Automated Network Request Management. The layered design connects standardized request intake with validation, approvals, fulfillment, notifications, tracking and audit capabilities while providing a structured foundation for security, scalability and future integrations.